

2

Lighting

In this chapter, we will learn about

- how different types of electric lights work
- the physical quantities for measuring brightness
- efficacy of electric lights

OXFORD UNIVERSITY PRESS

2.1 Electric lights

The electricity consumed for lighting in Hong Kong were about 10% of the total in 2020. This figure has continued to decrease over the past years (Fig 2.1a). The main reason is that the less energy-efficient types of lighting devices are gradually replaced by the more energy-efficient types. Nevertheless, the lighting devices all generate light by the same process triggered by various methods.

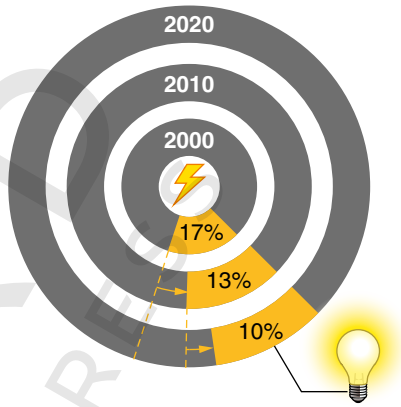


Fig 2.1a *The percentage of electricity consumption for lighting in Hong Kong. (Data obtained from EMSD website in Oct 2022)*

1 Light emission in electron transition

Cross-link
Refer to Book E2 Ch 2 for more details about the energy levels. ▶

All types of electric lights produce light by a process called *electron transition*, i.e. an electron in an atom moving from one **energy level** to another. Each energy level corresponds to the total energy that the electron can possess.

In general, an electron in an atom can only reside at one of the allowed energy levels. When it absorbs a specific amount of energy, it can jump to a higher energy level. The electron is said to be excited. The excited electron tends to jump to a lower energy level and energy is released (Fig 2.1b).

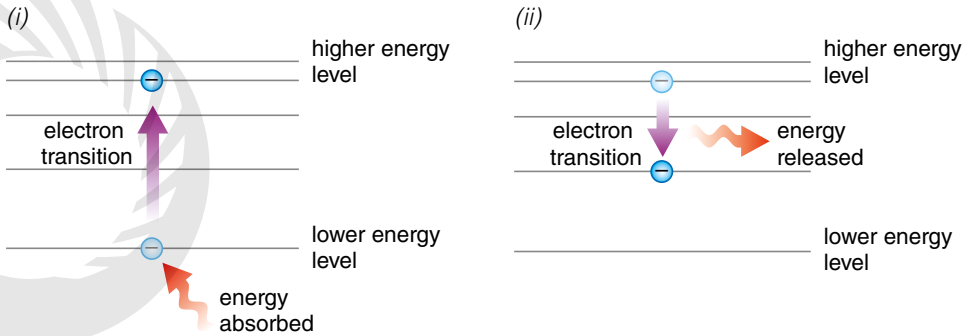


Fig 2.1b *Transition of electron to (i) a higher and (ii) a lower energy level.*

The energy released in an electron transition is in a form of electromagnetic (EM) wave. Visible light is given out only when the wavelength of the emitted EM wave lies in the visible range.

Know more

Energy and wavelength

If an electron transition release an EM wave of energy ΔE, the frequency f of the wave is given by:

$$\Delta E = hf$$

where h is called Planck constant. Its value is 6.63×10^{-34} J s.

ΔE is then related to the wavelength λ by:

$$\Delta E = h\left(\frac{c}{\lambda}\right) = \frac{hc}{\lambda}$$

where c is the speed of light in a vacuum.

electron transition 電子躍遷　　energy level 能級

DSE goal

State the different types of lighting used at home.

Describe how incandescent lamps, gas discharge lamps and light emitting diodes (LED) work and interpret light emission in terms of energy change in atomic level.

DSE exam

13(2)Q3.1, 18(2)Q.3(a)(i)

2 Common types of electric light

All electric lights emit visible light by electron transitions. We will look into three common types used at home. They are **incandescent lamp**, **gas discharge lamp** and **LED lamp**.

a Incandescent lamps (filament lamps)

The first electric light invented was the incandescent lamp. This type of lamp has a thin wire called a filament inside a glass globe filled with *inert gas* (Fig 2.1c). The filament is usually made of *tungsten* or a tungsten-based alloy. These materials have a high melting point and do not melt easily.

The melting point of tungsten is 3410 °C.

Inert gases such as *argon* and nitrogen are commonly used.

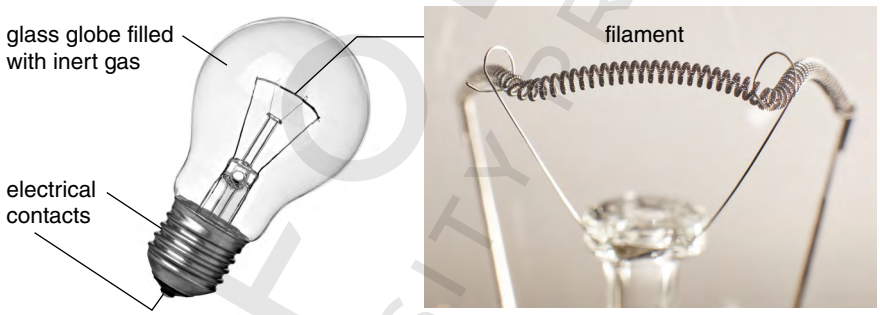


Fig 2.1c *The structure of an incandescent lamp.*

The following describes how light is generated in an incandescent lamp.

1 When a voltage is applied across the filament, an electric current flows through the filament and heats it to around 2500 °C.

2 At such a high temperature, the electrons in the filament are excited to different higher levels (Fig 2.1d).

3 When the electrons jump to many different lower energy levels, they release EM waves. Since the energy levels of the electrons in a solid metal are numerous and closely spaced, a continuous spectrum of EM waves is emitted (Fig 2.1e).

The mechanism is very much like what happens in the sun. This is why the light from incandescent lamps is said to appear 'natural'.

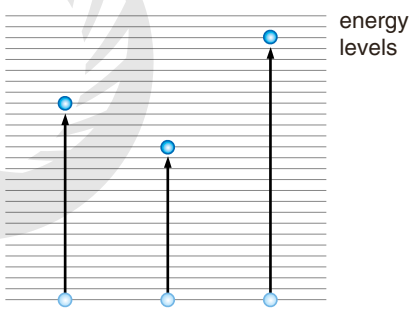


Fig 2.1d *Electrons are excited to different higher levels.*

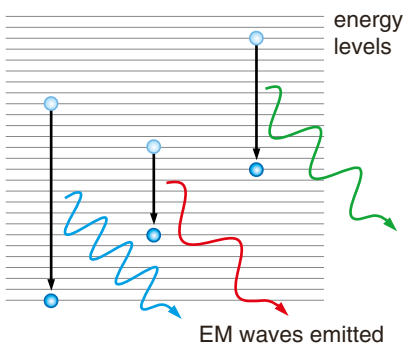


Fig 2.1e *Excited electrons jump to different lower energy levels and release EM waves of different frequencies.*

Know more

Invention of practical incandescent lamps

Do you know who invented the first practical incandescent lamp? *Thomas Edison*? Or somebody else?

▸ p.84

incandescent lamp 白熾燈　gas discharge lamp 氣體放電燈　LED lamp 發光二極管燈 / LED 燈
inert gas 惰性氣體　tungsten 鎢　argon 氬氣　Thomas Edison 愛迪生

An incandescent lamp is cheap, but it has two major drawbacks.

1 The lamp has a **very low energy efficiency** as only a small portion of energy is converted to light output. When the filament is heated to around 2500 °C, about 90–95% of its emission is not visible light, but mainly infra-red radiation (Fig 2.1f). Most of the energy is wasted in the form of heat.

Therefore, other than illumination, incandescent lamps are also used as heat lamps for warming.

Intensity is a measure of the amount of energy transferred through each unit of area per unit time.

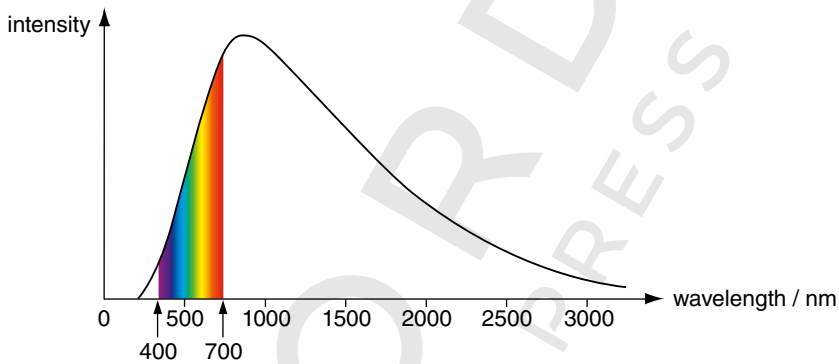


Fig 2.1f *Wavelengths of EM waves emitted by incandescent lamps.*

2 The lamp has a **short lifetime** as the filament blows easily (Fig 2.1g).

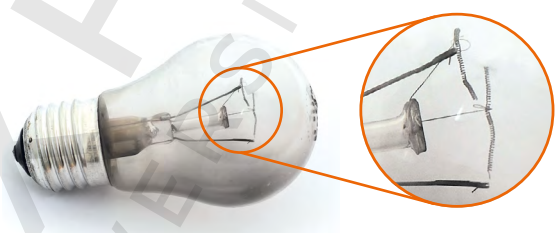


Fig 2.1g *A blown filament.*

Cross-link

Refer to Book 4 Ch 2 for more details about how the thickness of a conductor is related to its resistance and hence its temperature.

The heating effect of electric current make the filament hot, and evaporation occurs. Any thinner part of the filament becomes even hotter, and so evaporation occurs faster there (Fig 2.1h). The thinner part gets thinner and thinner until it breaks eventually. Although the inert gas inside the glass globe can slow down the evaporation, an incandescent lamp still fails after repeated use.

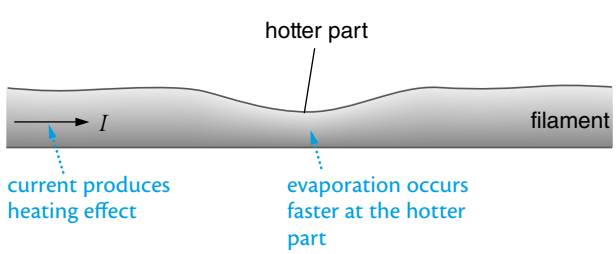


Fig 2.1h *Evaporation occurs faster at the thinner part of the filament.*

Incandescent lamps used for illumination are now being phased out in favour of more durable and energy-efficient lamps. Many places around the world have already restricted their sale.

Know more

The world's longest-lasting incandescent lamp

Believe it or not! There is an incandescent lamp that has been working for over one century!

▸ p.84

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Determine whether the following statements about electron transition are true.

(a) Visible light is always emitted when an electron jumps from a higher energy level to a lower energy level. (T/F)

(b) Energy is always emitted in an electron transition. (T/F)

2 How does an incandescent lamp produce light? Arrange the following steps in the correct order.

(1) Electric current heats up the filament.

(2) Electrons in the filament jump to lower energy levels to emit EM waves.

(3) Electrons in the filament are excited.

_____, _____, _____

3 An incandescent lamp emits EM waves of a broad range of frequencies because

A it does not produce light by electron transition.

B the energy levels of the electrons in the filament are numerous and closely spaced.

C it is not efficient in producing visible light.

D the filament is heated to a high temperature of around 2500 °C.

b Gas discharge lamps

In gas discharge lamps, electrons flow between two electrodes (discharge) through a gas. The **fluorescent tube lamp** (**FTL**) is a common type of gas discharge lamp (Fig 2.1i). This type of lamp has a sealed straight glass tube with contact pins at both ends.

A **compact fluorescent lamp** (**CFL**) is basically an FTL with twisted tube (Fig 2.1j) and with the positions of the pin rearranged to fit into a traditional lamp socket. It has a similar working principle as the FTL. We shall see how an FTL produces light.



Fig 2.1i A fluorescent tube lamp.



Fig 2.1j Compact fluorescent lamps.

fluorescent tube lamp (FTL) 熒光燈　　compact fluorescent lamp (CFL) 緊湊型熒光燈

Inside the sealed glass tube of an FTL, there are a small amount of mercury and an inert gas (e.g. argon) under a very low pressure. The inner wall of the tube has a *phosphor coating*. At each end of the tube, there is an electrode (Fig 2.1k).

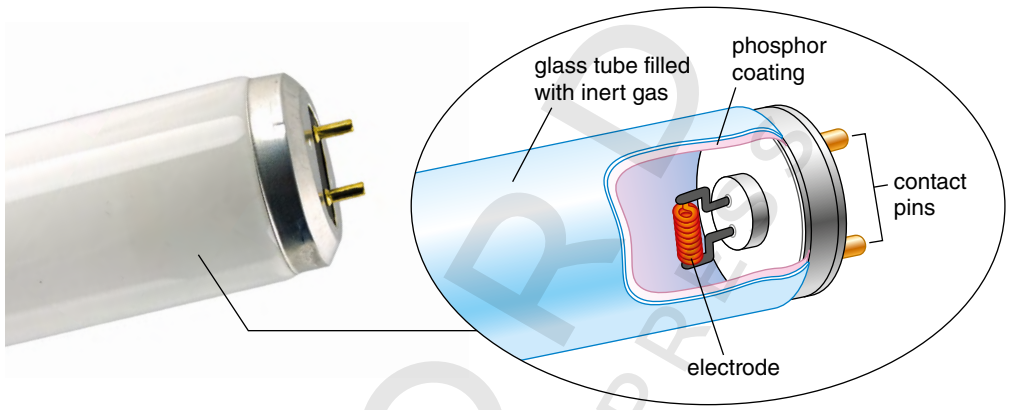
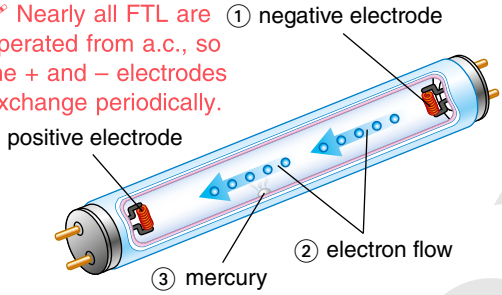


Fig 2.1k *An FTL and its components.*

→ This simulation shows the internal structure and the process of generation of light of an FTL. It is available on OUP web in PC version (.exe) only.

Simulation

The process of the generation of light in an FTL are as follows.



Nearly all FTL are operated from a.c., so the + and – electrodes exchange periodically.

Fig 2.1l *Electrons accelerated by a high voltage move along the tube.*

① When a high voltage is applied across the two electrodes, electrons are emitted from the negative electrode and accelerated by the high voltage.

② The electrons ionize the inert gas atoms to form ions and more free electrons. The ions and electrons are attracted towards the electrodes, giving rise to an electric current.

③ The electric current heats up the tube and causes the mercury inside the tube to vaporize.

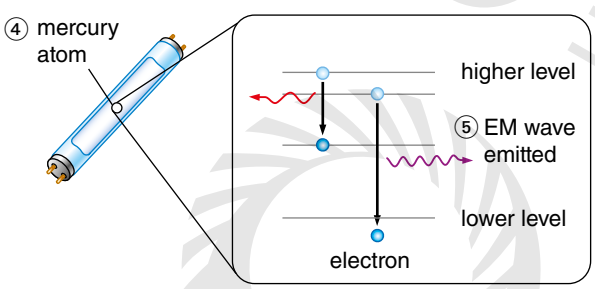


Fig 2.1m *Electron transitions in a mercury atom.*

④ Electrons and ions collide with the mercury atoms. As a result, the electrons in the mercury atoms are excited.

Remind Ss so that they do not mix up the electrons in the mercury atoms and the electrons producing the current.

⑤ The excited electrons in the mercury atoms jump to lower energy levels. Energy is released mainly in the form of ultra-violet (UV) radiation.

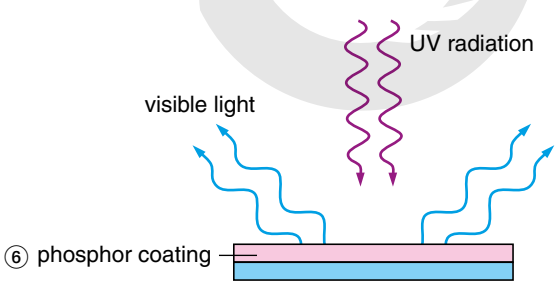


Fig 2.1n *Fluorescence of phosphor coating.*

⑥ The electrons in the phosphor coating absorb the ultra-violet radiation and are excited. When they jump to lower energy levels, a broad range of visible light is emitted for illumination. This process is called *fluorescence*.

phosphor coating 磷光質塗層　　fluorescence 熒光反應

FTLs and CFLs are efficient in producing light. However, they have some drawbacks.

1 They emit a small amount of ultra-violet radiation which may have adverse effects on health.

2 They contain mercury which is extremely toxic. Great care must be taken in disposal or when they are broken.

3 They need 10 to 30 seconds to reach their maximum brightness.

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 How does an FTL produce light? Arrange the following steps in the correct order.

i The mercury vaporizes.

ii Electron transitions occur in the mercury atoms to emit EM waves.

iii The phosphor coating emits visible light.

iv A high voltage is applied across the electrodes of the lamp.

v The phosphor coating absorbs ultra-violet radiation.

Know more

Fluorescent Lamp Recycling Programme

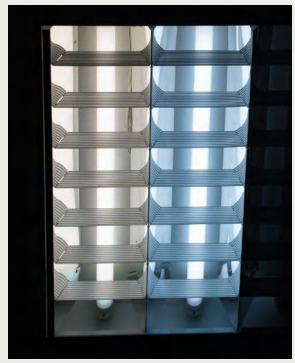
Spent FTLs and CFLs contain mercury, they should **not** be thrown away in rubbish bins. How should we dispose them?

▸ p.55



Phosphor coating

Phosphor coating is essential for FTLs and CFLs. If it is missing, these lamps would mainly emit ultra-violet radiation, which cannot be used for illumination. Moreover, the colour of the light emitted is determined by the phosphor coating. The coating actually contains three kinds of phosphor, each responsible for emitting one of the primary colours (red, green or blue) of light after absorbing ultra-violet radiation. The resulting light is perceived as white light. FTLs and CFLs are usually available in two kinds of white light, the yellowish white (called warm white) and the bluish white (called cool white). They are obtained by adjusting the proportion of the three phosphors.



The FTL on the left gives out warm white light and that on the right gives out cool white light.

OXFORD UNIVERSITY PRESS

Flipped classroom

Watch a video and answer the questions.

C LED lamps

LED lamps are rapidly gaining popularity across the world. An LED lamp contains a number of **light emitting diodes** (**LED**) (Fig 2.1o). They are available in light bulb shapes to fit in traditional lamp sockets or in tube shapes to fit in FTL sockets.



Fig 2.1o *(i) LED bulbs and (ii) LED tubes.*

An LED has two special materials sandwiched together. The two layers of materials are made from n-type **semiconductor** (the n-layer) and p-type semiconductor (the p-layer) (Fig 2.1p). In an n-type semiconductor, negative charge carriers (electrons) are in the majority. Conversely, in a p-type semiconductor, positive charge carriers (holes) are in the majority.

Note that both the n-type and p-type semiconductors are neutral in charge.

A semiconductor has electrical conductivity between a conductor and an insulator. Its conductivity can be varied by adding controlled impurities to it. Silicon is commonly used to made semiconductors commercially. However, in making LEDs, other materials are used instead.

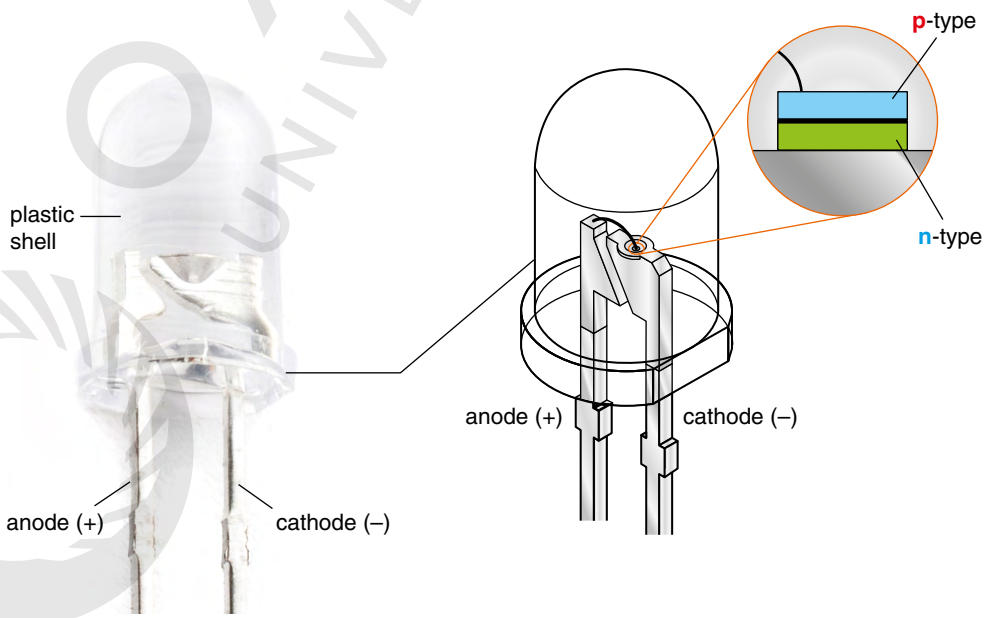
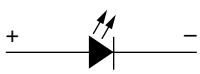


Fig 2.1p *The structure of an LED.*

The circuit symbol of an LED is:



A current can only flow in the direction indicated by the big arrowhead.

A current can only flow through an LED in one direction from anode to cathode. A circuit for converting a.c. to d.c. is included in an LED lamp so that it can be operated normally by mains electricity.

light emitting diode (LED) 發光二極管　　semiconductor 半導體

→ This simulation shows how electrons and holes move in an LED.
It is available on OUP web in PC version (.exe) only.

Simulation

The forward voltage refers to the amount of voltage required for the current to flow through the p-n junction. Therefore, the voltage at the anode should be more positive than that at the cathode.

The following describes how light is generated in an LED.

▶ ① When a forward d.c. voltage is applied to an LED, electrons and holes move towards the boundary called the p-n junction between the p-type and n-type semiconductors (Fig 2.1q).

② An electron will combine with a hole near the p-n junction. When this happens, the electron jumps from a higher to a lower energy level. The excess energy is released in the form of an EM wave (Fig 2.1r).

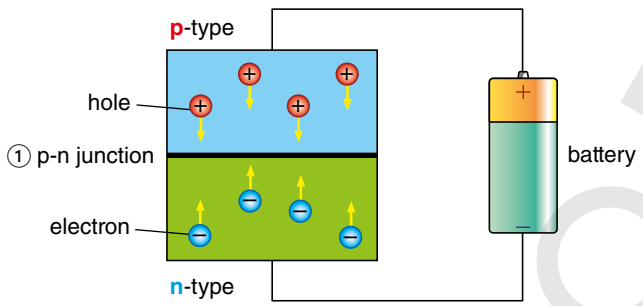


Fig 2.1q *An LED connected to a battery.*

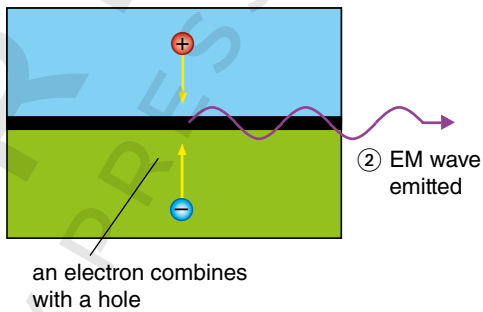


Fig 2.1r *An electron combines with a hole and releases the excess energy as an EM wave.*

The light emitted by an LED is within a narrow spectrum and can be regarded as monochromatic. The colour of the light is determined by the energy released during the electron-hole combination, which in turn is determined by the materials forming the p-n junction.

Since an LED cannot produce white light directly, an LED lamp produces white light either by combining blue, green and red LEDs (Fig 2.1s) or using a phosphor coating to convert blue light to white light.

LEDs can be designed to emit only visible light, so it is an efficient light source.

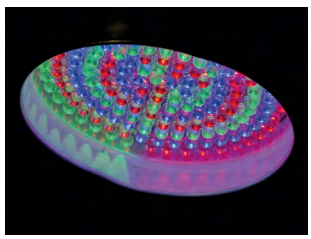


Fig 2.1s *An LED lamp with red, green and blue LEDs.*

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 For an LED to work properly, the p-layer should be connected to the (positive/negative) terminal and the n-layer to the (positive/negative) terminal of a battery. The (electrons/holes) in the p-layer and the (electrons/holes) in the n-layer move towards the p-n junction and combine to emit EM waves.

2 The frequency of the light emitted by an LED depends on

- ☐ the material forming the p-layer and n-layer.
- ☐ the temperature of the LED.
- ☐ the voltage across the LED.

DSE goal

Discuss cost effectiveness of incandescent lamps, gas discharge lamps and light emitting diodes.

DSE exam

17(2)Q.3(c)

3 Cost effectiveness

Table 2.1a compares the three types of electric light.

	Incandescent lamp 	CFL 	LED lamp 
Price	☺ low	☹ high	☹ high
Lifetime	☹ short (750–4000 h)	☺ long (7000–24 000 h)	☺ long (15 000–50 000 h)
Efficiency	☹ low (about 90–95% of emission are not visible light)	☺ high	☺ high
Environment impact	☺ contains no mercury	☹ contains mercury	☺ contains no mercury
Heat released	☹ much	☺ some	☺ some
UV radiation	😐 yes (little, emissions are much lower than CFLs)	☹ yes	☺ no (for some colours of LED lamps)

Table 2.1a *Comparison of different types of electric light.*

The lifetime of an LED lamp is established on a prediction of when the light output is decreased by 30%.

A longer lifetime may also help reduce the labour cost for replacement.

CFLs and LED lamps are more expensive than incandescent lamps, but they can save money in the long run as they have longer lifetime and higher energy efficiency. See Example 1 below.

Example 1 Cost effectiveness of lamps

The information of an LED lamp, a compact fluorescent lamp (CFL) and an incandescent lamp are shown below. The three lamps have the same brightness.

	LED lamp	CFL	Incandescent lamp
Power	10 W	14 W	60 W
Lifetime	15 000 hours	10 000 hours	1000 hours
Cost	$50	$40	$15

Suppose the cost of electricity is $1.1 per kW h.

(a) Which lamp is the cheapest for 1000 hours of use?

(b) Which lamp is the cheapest for 15 000 hours of use?

(c) What are the advantages of using LED lamp over the other two lamps?

Solution

Tip
Remember to convert the power of lamp from W to kW.

(a) Cost of using LED lamp = $1.1 \times 0.01 \times 1000 + 50 = \$61$
Cost of using CFL = $1.1 \times 0.014 \times 1000 + 40 = \$55.4$
Cost of using incandescent lamp = $1.1 \times 0.06 \times 1000 + 15 = \$81$
$\therefore$ CFL is the cheapest for 1000 hours of use.

Tip
More than one CFL and incandescent lamp are needed for 15 000 hours of use.

(b) Cost of using LED lamp = $1.1 \times 0.01 \times 15\,000 + 50 = \$215$
Cost of using CFL = $1.1 \times 0.014 \times 15\,000 + 2 \times 40 = \$311$
Cost of using incandescent lamp = $1.1 \times 0.06 \times 15\,000 + 15 \times 15 = \$1215$
$\therefore$ LED lamp is the cheapest for 15 000 hours of use.

(c) LED lamp has a longer lifetime and a higher energy efficiency.

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 Consider an LED lamp, a gas discharge lamp and an incandescent lamp of the same brightness. Determine whether the following statements are true.

(a) The gas discharge lamp has the highest input power. (T/F)

(b) The cost of electricity of using the LED lamp is the lowest. (T/F)

(c) The power rating of the incandescent lamp is the highest. (T/F)

Learn by practice 1

1 A 18-W CFL is replaced by a 13-W LED lamp of the same brightness. The prices of the CFL and the LED bulb are \$35 and \$55 respectively. Both lamps have lifetime over 10 000 hours. The cost of electricity is \$1.1 per kW h. The lamp is used for 5 hours every day.

(a) How many days will it take for the difference in the cost of electricity to compensate for the difference in price between the two lamps?

(b) Find the total cost that will be saved after 3 years.
Take 1 year = 365 days.

Know more

Fluorescent Lamp Recycling Programme (p.51)

In Hong Kong, the Fluorescent Lamp Recycling Programme (FLRP) launched by the Environmental Protection Department provides collection boxes for the general public to discard mercury-containing lamps for recycling. You just need to place the spent lamps in the packaging that came with new lamps and put them in the collection box. Have you ever seen any of the FLRP collection boxes?



Exercise 2.1

1 Which of the following is **not** an advantage of an LED lamp over an incandescent lamp (Fig a)?



Fig a

A Higher energy efficiency
B Longer lifetime
C Lower selling price
D Lower operating cost

2 When a fluorescent tube lamp operates, mercury is vaporized and is then bombarded by electrons and ions (Fig b).

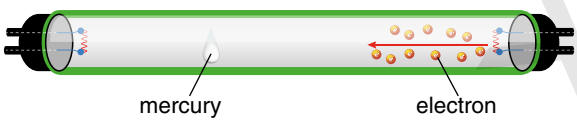


Fig b

In consequence, the mercury atoms
A give out mainly visible light.
B give out mainly ultra-violet radiation.
C give out mainly electrons.
D give out mainly fluorescent materials.

***3** Stacy buys lamp *P* of power rating 50 W from a shop. The lamp is used for 8 hours a day and the cost of electricity is $1.1 per kW h. Lamp *Q* is rated at 30 W and is $20 more expensive than lamp *P*. If she bought lamp *Q*, how much money would be saved in 365 days?
A $38.4
B $44.2
C $58.2
D $64.2

***4** Which of the following statements about an LED lamp (Fig c) is/are correct?



Fig c

(1) It contains mercury.
(2) It can be operated by mains electricity.
(3) It has a longer lifetime than fluorescent tube lamps and incandescent lamps.
A (1) only
B (3) only
C (2) and (3) only
D (1), (2) and (3)

***5** (a) Explain, in terms of electron transition, how an electric lamp gives out light.
(b) Suggest a possible source of energy loss by an electric lamp.

***6** Suggest why an incandescent lamp has a much lower energy efficiency than an LED lamp.

***7** What would happen to the light emitted by a fluorescent tube lamp if
(a) the phosphor powder was detached from the inner wall of the tube;
(b) there was no mercury inside the tube.

***8** Light is produced by electron transitions in different types of electric lights. Describe how the following electric lights trigger electron transitions to produce light.
(a) Fluorescent tube lamp
(b) Incandescent lamp
(c) LED lamp

OXFORD UNIVERSITY PRESS

2.2 Measuring brightness

Suppose there are two lamps emitting red and green light of the same power (Fig 2.2a). Which one will look brighter?



Fig 2.2a *Lamps emitting red and green light of the same power.*

The answer is the green light. This is because the human eye is most sensitive to green-yellow light (λ ~555 nm) (Fig 2.2b). Visible light of other wavelengths has to be emitted at a higher power to give the brightness identical to 555-nm light as perceived by the human eye.

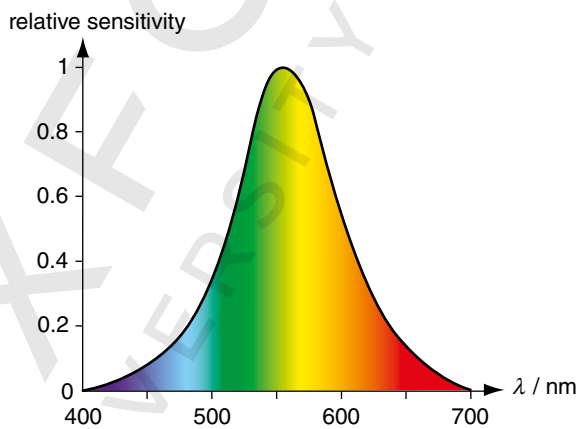


Fig 2.2b *Sensitivity of the human eye to different wavelengths of visible light.*

1 Quantities related to illumination

a Luminous flux

DSE goal

Realise that the eye response depends on wavelengths.

Define luminous flux as the energy of light emitted per unit time by a light source.

Use lumen as a unit of luminous flux.

DSE exam

12(2)Q3.1, 14(2)Q3.1, 18(2)Q.3(a)(ii)

To measure how bright a light source is to the human eye, **luminous flux** is used.

Luminous flux is the power output of a light source corrected for the dependence on wavelength of the sensitivity of the human eye.

▶ Two light sources of the same luminous flux show the same brightness to us regardless of their colours.

Luminous flux is measured in **lumens**, written as **lm**. The symbol for luminous flux is Φ. Note that only the power of **visible light** is included in calculating the luminous flux.

luminous flux 光通量　　lumen (lm) 流明

✎ Remind Ss that watt and lumen are dimensionally the same but cannot be substituted for each other. Whenever we use lumen, the power has been adjusted according to the human eye's sensitivity.

A light source consuming a large amount of energy is not necessarily a 'bright' source. Therefore, when we want to choose an electric light for illumination, the luminous flux is more useful than the power rating (Fig 2.2c).



Fig 2.2c *The package of a 20-W CFL.*

✎ The luminous flux of a standard 100-W incandescent lamp is about 1600 lm. According to Fig 2.2c,
rated power of CFL = 20 W
Φ of CFL = 70 × 20 = 1400 lm
Therefore, this CFL is actually slightly dimmer than a standard 100-W incandescent lamp.

Figure 2.2d shows the typical luminous flux of some light sources.

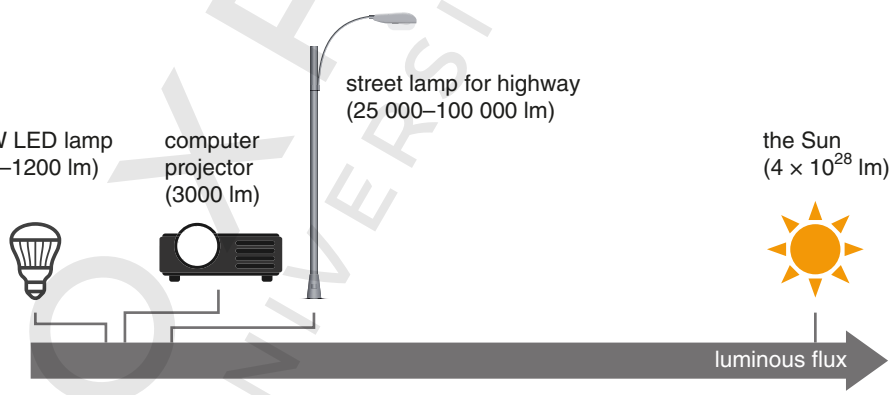


Fig 2.2d *Luminous flux of some light sources.*

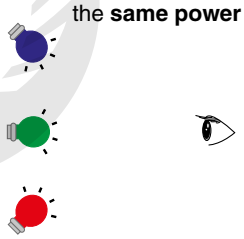
Google Forms Microsoft Forms

Quick check 5

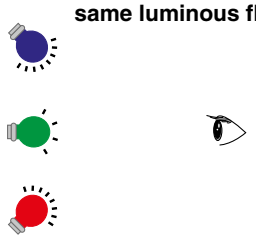
→ E-version in OUP Exercise Platform is also available.

1 In each of the following cases, which light source looks the brightest?

(a) Light sources emitting light of the **same power**



(b) Light sources of the **same luminous flux**



2 Light bulbs *P* and *Q* have the same size and emit visible light of the same power. *P* emits blue light while *Q* emits yellow light. Which light bulb looks brighter?

DSE goal

Define illuminance as luminous flux falling on unit area of a surface.

Use lux as a unit of illuminance.

Use inverse square law and Lambert's cosine law to solve problems involving illuminance.

DSE exam

13(2)Q.3(a), 15(2)Q3.1, 16(2)Q3.1, Q3.2, 17(2)Q3.1, 19(2)Q3.1, 20(2)Q3.1, 22(2)Q3.2, 23(2)Q3.2

b Illuminance

i Basic understanding

Luminous flux describes how bright a light source is. However, not all the light emitted from a light source can reach the object to be illuminated. To describe how brightly a surface is illuminated, we can use **illuminance**, denoted by E. It is the luminous flux falling on a surface per unit area.

$$E = \frac{\Phi}{A}$$

(E: illuminance (lx); Φ: luminous flux (lm); A: area (m^2))

Illuminance is measured in **lux**, written as **lx**.

From the above equation,

$$1 \text{ lx} = 1 \text{ lm m}^{-2}$$

Illuminance is a measurement of the **power received** per unit area. It can be measured with a *light meter* (Fig 2.2e). This device is commonly used by photographers (Fig 2.2f).

You can use your smartphone as a light meter with a suitable app.

Useful link

iOS Android



Fig 2.2e A light meter.



Fig 2.2f A photographer using a light meter.

Space for various functions may need different illuminances. For example, a room for reading or normal working may need 500–1000 lx, while a corridor may require only 100 lx. With such information, you can have a general idea on what lamp to choose for a place.

The illuminance under direct sunlight at midday could be as high as 100 000 lx! Therefore, reading under direct sunlight should be avoided.

Luminous flux measures the power emitted by a source while illuminance measures the power per unit area received by a surface, both corrected for the dependence on wavelength of the sensitivity of the human eye.

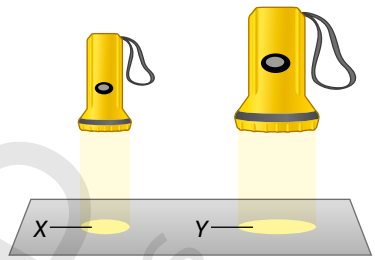
illuminance 照明度 / 光照度　　lux (lx) 勒克司　　light meter 測光儀

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 Two torches of the same luminous flux are shone towards a surface as shown. Which region, *X* or *Y*, has a higher illuminance?



ii Light falling at an angle

Suppose light from a parallel light source with a luminous flux Φ falls perpendicularly on a surface, illuminating an area $A_\perp$ (Fig 2.2g(i)). The illuminance $E_\perp$ at that area is

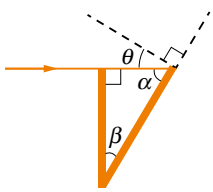
$$E_\perp = \frac{\Phi}{A_\perp}$$

If the parallel light makes an angle θ with the normal of the surface, the light will fall on a larger area A (Fig 2.2g(ii)). Therefore,

$$E = \frac{\Phi}{A} = \frac{\Phi}{\frac{A_\perp}{\cos\theta}} = \left(\frac{\Phi}{A_\perp}\right)\cos\theta = E_\perp \cos\theta$$

This relation is often referred to as **Lambert's cosine law**.

Tip



$\alpha + \theta + 90° = 180°$
$\alpha + \beta + 90° = 180°$
$\therefore \beta = \theta$

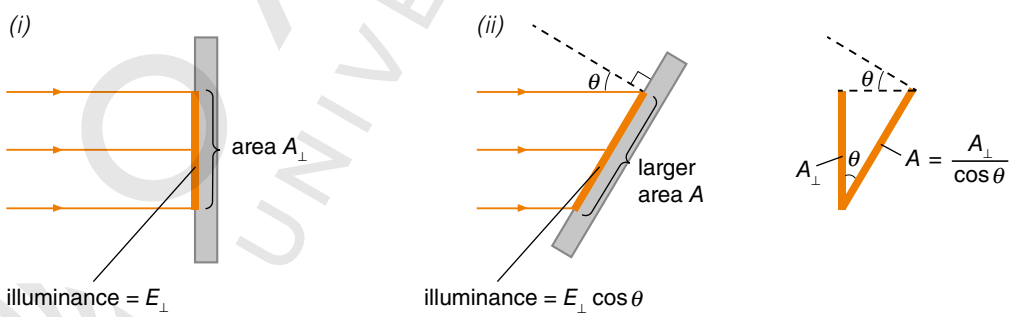


Fig 2.2g *Parallel light falling on a surface (i) perpendicularly and (ii) at an angle.*

Example 2 Parallel light beam

A torch emitting a parallel light beam has a luminous flux of 10 lm. The light beam falls perpendicularly on a plane surface (Fig a). A light meter is placed in the illuminated region. The illuminance it measures is 400 lx.

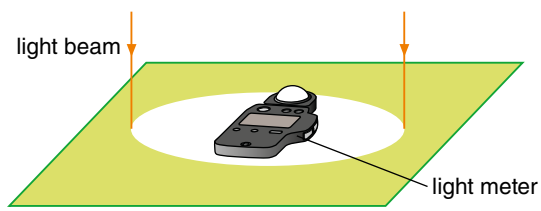


Fig a

Lambert's cosine law 朗伯餘弦定律

(a) Find the area of the illuminated region.

(b) The torch is then tilted and the light meter reading becomes 332 lx (Fig b). What is the angle θ between the light beam and the normal of the surface?

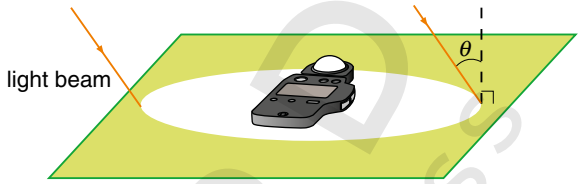


Fig b

Solution

(a) By $E = \frac{\Phi}{A}$,

$$A = \frac{\Phi}{E}$$

$$= \frac{10}{400}$$

$$= 0.025 \text{ m}^2$$

(b) By $E = E_{\perp}\cos\theta$,

$$332 = 400 \cos\theta$$

$$\theta = 33.9°$$

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

True or false:

1 When a parallel light beam shines on a surface, the illuminance on the surface is maximum if the surface is normal to the beam. (T/F)

2 When a parallel light beam shines on a surface, the illuminance on the surface becomes larger if the light source is brought closer. (T/F)

Learn by practice 2

1 Sam shines a parallel light beam from a torch on a plane surface. The luminous flux of the torch is 40 lm. When the light beam falls perpendicularly on the surface, it illuminates an area of 20 cm^2.

(a) Find the illuminance at the illuminated region.

(b) Sam now shines the light beam at 30° to the normal of the surface. Find the new illuminance at the illuminated region using the result in (a).

OXFORD UNIVERSITY PRESS

iii Point source of light

Have you ever used a spray? If you spray some paint onto a piece of paper from a larger distance, you will get a larger patch of colour. However, the colour will be lighter, i.e. the amount of paint per unit area is smaller (Fig 2.2h).

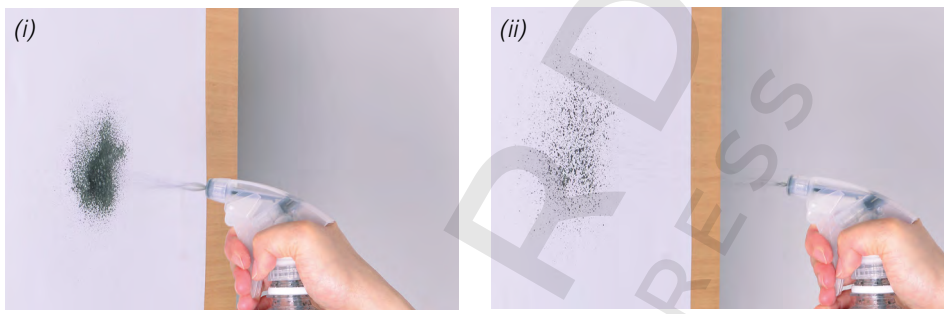


Fig 2.2h *Applying spray at (i) a smaller distance and (ii) a larger distance.*

✎ It is impossible for a light source to be perfectly uniform in all directions (isotropic). However, if r is great enough, it is an accurate approximation.

The light emitted from a point source, which emits light evenly in all directions, behaves similarly as the paint from the spray. Consider a point source S. If we draw a sphere centred at S, the light from S will spread evenly on the surface of the sphere (Fig 2.2i). As the light from S travels further, it spreads across a larger spherical surface. Therefore, the light received per unit area, hence the illuminance, decreases (Fig 2.2j).

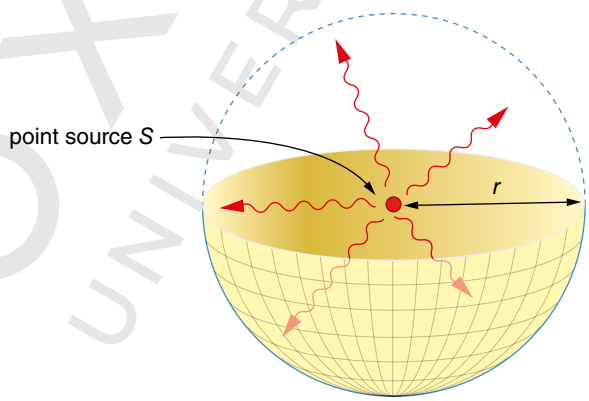


Fig 2.2i *Light from a point source spreads evenly on a spherical surface.*



→ This is a Video Quiz about an experiment verifying the inverse-square law of intensity.

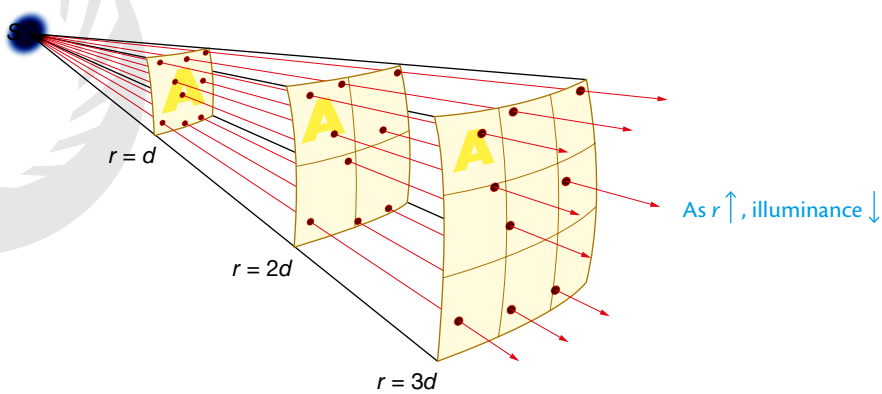


Fig 2.2j *The illuminance (indicated by the density of red dots) decreases as the distance from the source increases.*

A spherical surface of radius r has an area of $4\pi r^2$. If the luminous flux of the source is Φ, the illuminance on the surface will be

$$E_\perp = \frac{\Phi}{A} = \frac{\Phi}{4\pi r^2}$$

Therefore, if the light from point source S falls perpendicular on point X on a surface (Fig 2.2k), the illuminance at X is $\frac{\Phi}{4\pi r^2}$.

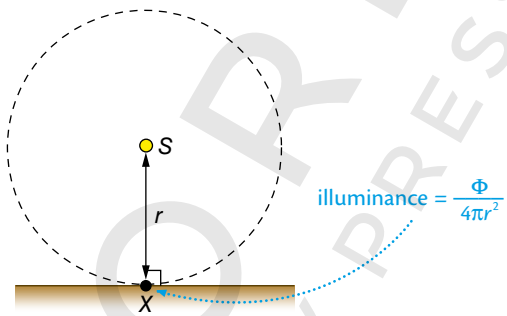


Fig 2.2k *Light from a point source falls perpendicularly on a point on a surface.*

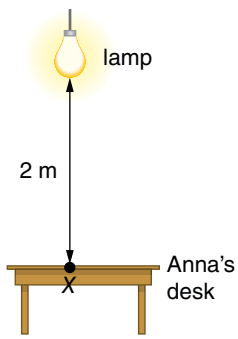
We can see that the relation between illuminance and distance obeys the **inverse-square law**:

$$E_\perp \propto \frac{1}{r^2}$$

This law only holds for a point source. For a parallel light source, the illuminance is independent of the distance from the source.

Example 3 Illuminance on the desk (perpendicularly)

A lamp of luminous flux 7500 lm is 2 m directly above point X on Anna's desk. Find the illuminance at point X. Assume that the lamp is small enough to be regarded as a point source and that reflection is negligible.



Solution

Illuminance $E_\perp = \frac{\Phi}{4\pi r^2} = \frac{7500}{4\pi \times 2^2} = 149$ lx

inverse-square law 平方反比定律

→ This video demonstrates the Phone experiment.

Video

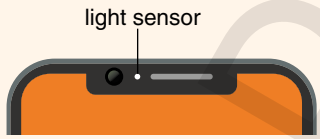
Phone experiment

→ Worksheet is available on OUP web.

Verifying the inverse-square law of illuminance

The light sensor is used to adjust the brightness of the screen automatically according to the environment.

Smartphones are usually equipped with a light sensor. The sensor is usually located at the top of the screen.



What to do

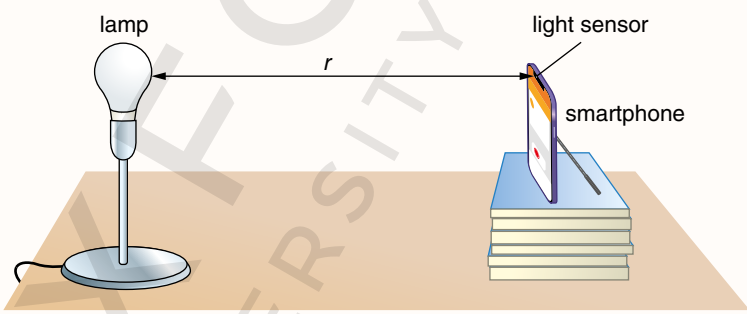
You may search the app on the Internet.

light meter app

1 Open a light meter app that allows your smartphone to measure illuminance.

2 Place a lamp at a distance r away from the light sensor. Ensure the sensor faces the light bulb normally so that the sensor aligns with the light bulb.

Remove any lamp shade.



3 Darken the room and switch on the lamp. Record the readings E of the phone for different values of r.

4 Calculate the values of $\frac{1}{r^2}$. Plot a graph of E against $\frac{1}{r^2}$.

What you find

What is the relationship between E and $\frac{1}{r^2}$?

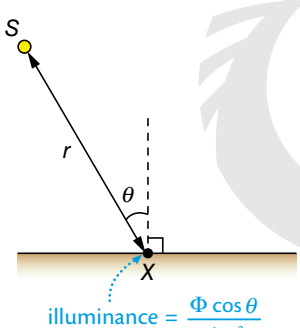


Fig 2.2l Light from a point source hits point X on a plane surface at an angle θ.

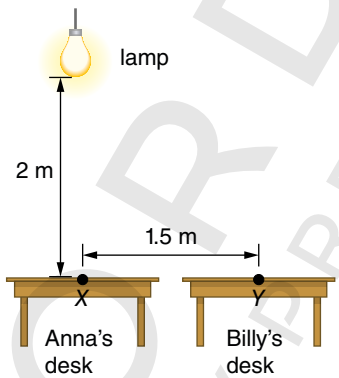
How should we calculate the illuminance if the light from a point source falls at an angle on a plane surface? Suppose the light hits point X on the surface at an angle θ (Fig 2.2l). By Lambert's cosine law, the illuminance at X is given by

$$E = E_{\perp} \cos\theta = \frac{\Phi \cos\theta}{4\pi r^2}$$

E: **illuminance** (lx); Φ: **luminous flux** (lm); θ: **angle** between light and normal of surface; r: **distance** from the light source (m)

Example 4 | Illuminance on the desk (at an angle)

A lamp of luminous flux 7500 lm is 2 m directly above point X on Anna's desk. Find the illuminance at point Y on Billy's desk, where $XY = 1.5$ m. Assume that the lamp is small enough to be regarded as a point source and that reflection is negligible.

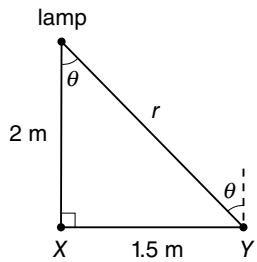


Solution

$$r = \sqrt{2^2 + 1.5^2} = 2.5 \text{ m}$$

$$\cos\theta = \frac{2}{2.5}$$

$$E = \frac{\Phi \cos\theta}{4\pi r^2} = \frac{7500 \times \frac{2}{2.5}}{4\pi \times 2.5^2} = 76.4 \text{ lx}$$



Know more

Reflection of light by walls

For simplicity, we have ignored the reflection of light from the ceiling and walls. However, this factor contributes significantly to the illuminance in practice. A white wall could reflect as much as 80–85% of light incident on it. Painting a room white can make it bright!



Google Forms Microsoft Forms

Quick check 8

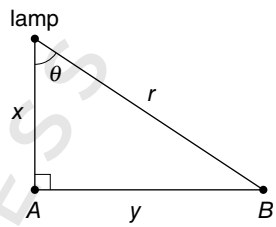
→ E-version in OUP Exercise Platform is also available.

True or False:

1 Inverse-square law only applies to point light sources. (T/F)

2 For a point light source, illuminance is independent of the distance from the source. (T/F)

3 Consider the setting as shown. Suppose the lamp is a point light source. Complete the table below.



	Φ of lamp	x	y	θ	r	E at A	E at B
(a)	880 lm	3 m	4 m	53.1°			
(b)			2 m		4 m	12.8 lx	

Learn by practice 3

1 A pendant lamp is installed in a room (Fig a). It can be regarded as a point light source and has a luminous flux of 400 lm. P and Q are points on the floor and the wall in the room. The light from the lamp falls perpendicularly at both P and Q.

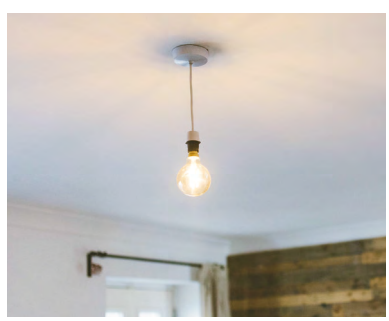


Fig a

(a) P is at 3 m from the lamp. Find the illuminance at P.

(b) The distance of Q from the lamp is 2.5 times the distance of P from the lamp. Find the illuminance at Q using the result in (a).

2 A point source of light is placed at P and the illuminance at X is 38 lx (Fig b). Find the luminous flux of the light source.

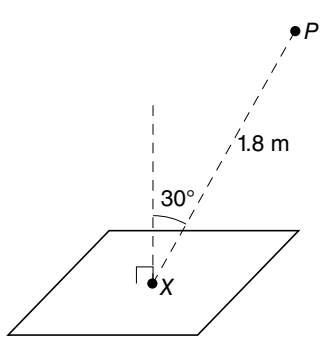


Fig b

Now, we have learned how to find the illuminance in different cases. Table 2.2a helps us determine which equation should be used to find the illuminance at a point (X) in each case.

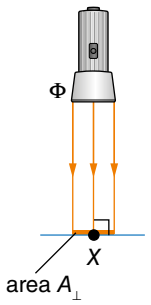
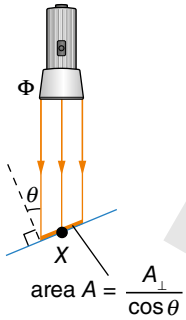
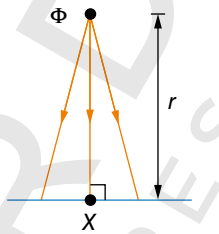
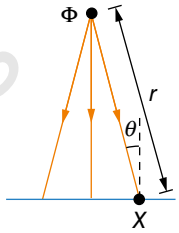
Parallel source		Point source	
			
$E_\perp = \frac{\Phi}{A_\perp}$	$E = E_\perp \cos\theta = \frac{\Phi \cos\theta}{A_\perp}$	$E_\perp = \frac{\Phi}{4\pi r^2}$	$E = E_\perp \cos\theta = \frac{\Phi \cos\theta}{4\pi r^2}$

Table 2.2a *Illuminance E at point X on a surface in different cases.*

More information about light pollution in Hong Kong can be found at:

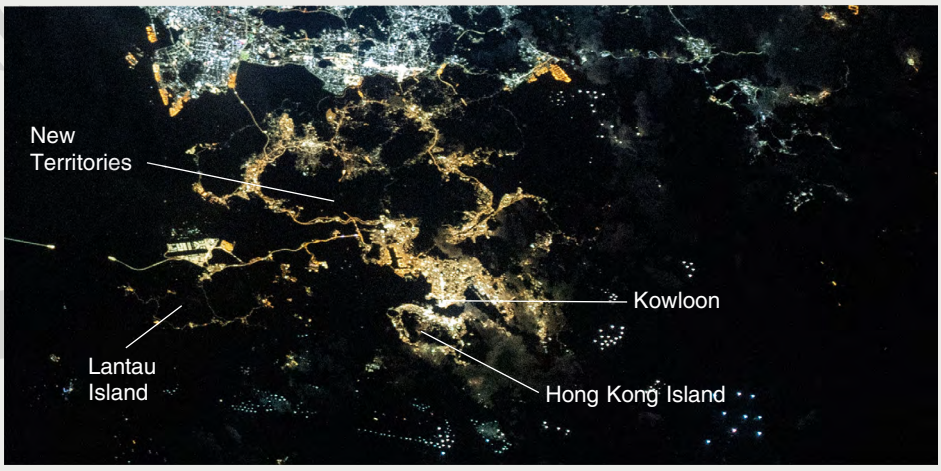
STSE

Light pollution

Hong Kong was once named the world's worst city for light pollution. The unnecessary electric lighting wastes electricity; the excessive brightness may also damage the health of people and wildlife.

In a remote place, the illuminance at night is lower than 1 lux. However, on some streets in Hong Kong, the value can be as high as 9000 lux!

Are there any regulations to minimize light pollution? What can you do to improve the situation?



Nighttime picture of Hong Kong as seen from the International Space Station in space.

☆ HKDSE 2015 Paper 2 Q3.1

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Inverse-square law and Lambert's cosine law

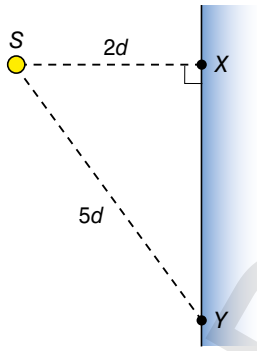


Fig a

A point source of light S illuminates a wall as shown (Fig a). Points X and Y on the wall are at distances $2d$ and $5d$ from S respectively. If the illuminance at X is 500 lx, what is the illuminance at Y?

A 12.8 lx
B 32 lx
C 73.3 lx
D 80 lx

Solution

By $E_\perp \propto \frac{1}{r^2}$,

$$\frac{E_{Y\perp}}{E_{X\perp}} = \frac{\frac{1}{r_Y^2}}{\frac{1}{r_X^2}} = \frac{r_X^2}{r_Y^2} = \frac{(2d)^2}{(5d)^2} = \frac{4}{25}$$

$$\Rightarrow E_{Y\perp} = \frac{4}{25}E_{X\perp} = \frac{4}{25} \times 500 = 80 \text{ lx}$$

Applying inverse-square law and Lambert's cosine law separately can avoid more complicated expression during calculation. But bear in mind that the result obtained here is the illuminance at Y if the incident light hit the wall at Y **perpendicularly**.

Let θ be the angle between the incident light and the normal at Y (Fig b).

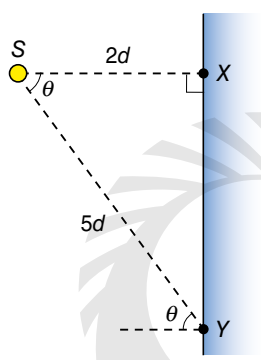


Fig b

$$\cos\theta = \frac{2d}{5d} = \frac{2}{5}$$

Common mistake ⚠
Students may employ wrong angle or get sine and cosine mixed up. The angle θ should be the one between the incident light and the normal of the wall at Y.

Apply Lambert's cosine law.

$$E_Y = E_{Y\perp}\cos\theta = 80 \times \frac{2}{5} = 32 \text{ lx}$$

∴ The answer is B.

▶ Revision 2 Q12 (p.79)

OXFORD UNIVERSITY PRESS

DSE goal

Define efficacy of electric lights as a ratio of luminous flux (lm) to electrical power input (W) and solve related problems.

DSE exam

17(2)Q3.2, 18(2)Q.3(b), 21(2)Q3.1, 22(2)Q3.1, 23(2)Q3.1

The calculation of efficiency includes the power of emission of all EM waves and does not take the sensitivity of the human eye into account, so it is not a good measurement of the effectiveness of a lamp.

Experiment 2a

Comparing the efficacy of different types of light sources (p.188)

Video



→ This video demonstrates Experiment 2a.

2 Efficacy

Two different electric lights may have the same luminous flux but consume different power. **Luminous efficacy**, or simply **efficacy**, measures how much electrical energy is converted into visible light, corrected for the dependence on wavelength of the sensitivity of the human eye.

$$\text{Efficacy} = \frac{\text{luminous flux}}{\text{input power}}$$

in lm W^{-1} · in lm · **rated power** of light (W)

It is measured in lm W^{-1}.

Even the same type of light may have a wide range of efficacy. In general, LED lamps, FTLs and CFLs have much higher efficacy than incandescent lamps (Table 2.2b).

		Incandescent lamp	CFL	LED lamp
				
Luminous flux	450 lm	40 W	9 W	7 W
	800 lm	60 W	14 W	10 W
	1100 lm	75 W	19 W	17 W
	1600 lm	100 W	23 W	20 W
Efficacy		**10–20 lm W^{-1}**	**30–100 lm W^{-1}**	**30–100 lm W^{-1}**

Table 2.2b *Some reference values for different types of electric lights.*

Example 5 Blue light and green light

Two 880-lm lamps, *A* and *B*, emit monochromatic light of 555-nm green light and 480-nm blue light respectively. The rated power of *A* is 40 W and that of *B* is 80 W.

(a) Compare the brightness of *A* and *B*. Explain briefly.

(b) Find the efficacy of each lamp.

(c) Is the green light or the blue light emitted at a higher power? Why?

luminous efficacy / efficacy 光視效能

Solution

(a) A and B have the same brightness, because they have the same luminous flux.

(b) Efficacy of $A = \frac{880}{40} = 22$ lm W^{-1}

Efficacy of $B = \frac{880}{80} = 11$ lm W^{-1}

(c) The blue light is emitted at a higher power. Since our eyes are less sensitive to blue light than green light, B needs to emit blue light at a higher power so that it produces the same luminous flux as A.

Example 6 Energy label on an LED lamp

The energy label affixed on an LED lamp (Fig a) and the information on its package (Fig b) are shown below.



Fig a

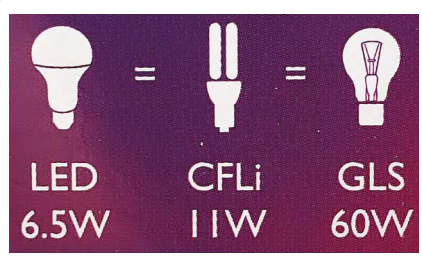


Fig b

(a) What is the luminous flux of the lamp?

(b) The package indicates that this LED lamp is as bright as a 60-W GLS lamp. What is the efficacy of the 60-W GLS lamp?

(c) Among the three lamps shown on the package, which lamp is the most energy-saving one? Explain your answer.

Solution

(a) Luminous flux = efficacy × input power
= 110 × 6.5
= 715 lm

(b) Efficacy $= \frac{715}{60} = 11.9$ lm W^{-1}

(c) The LED lamp is the most energy-saving lamp. This is because it consumes the least amount of power to give the same luminous flux.

Google Forms Microsoft Forms

Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) If two lamps having the same luminous flux give out the same colour of light, they must have the same efficacy. (T/F)

(b) If a green light source has the same efficacy as a red light source, the green light source must look brighter than the red light source. (T/F)

2 Fill in the table with the corresponding units.

Quantity	power rating	luminous flux	illuminance	efficacy
Unit				

Table a

3 The specification of three lamps *X*, *Y* and *Z* are given below (Table b).

	X	***Y***	***Z***
Power rating	13 W	75 W	
Luminous flux	920 lm		1600 lm
Efficacy		14 lm W^{-1}	80 lm W^{-1}

Table b

(a) Complete Table b.

(b) Which lamp is the best choice to buy in terms of efficacy?

Learn by practice 4

1 An energy-saving lamp has a luminous flux of 2700 lm and a power rating of 45 W. Find its efficacy.

2 The efficacy of a table lamp is 65 lm W^{-1} when it is operated at 15 W. What is its luminous flux?

3 An ideal light source of power 1 W emits 683 lumens of monochromatic green light at 555 nm. A white light source of power 1 W gives about one-tenth of this luminous flux.

(a) Find the luminous flux of the white light source.

(b) Find the efficacy of the white light source.

Know more

The maximum possible efficacy of a lamp

Suppose a lamp is ideal. It is 100% efficient and can convert all electrical energy to monochromatic light of wavelength of 555-nm (green-yellow light). In this case, the lamp has an efficacy of 683 lm W^{-1}. This is the maximum possible efficacy of any lamp since the human eye is most sensitive to light of wavelength 555 nm and a light source can never be more efficient than 100%.

DSE goal

Suggest examples of energy-saving devices.

3 Energy-saving devices

In Chapter 1 and this chapter, we have learned that the same type of household appliances may have different energy efficiency. Other than choosing energy-efficient appliances, we can also make use of energy-saving devices to help save energy at home. The following suggests some examples.



Timer

It switches an appliance on and off at pre-set times.



Dimmer for lighting

It controls the brightness of an electric light.



Pressure cooker

This cooking vessel can trap steam to build up a high pressure. This allows the water and the food inside to reach temperatures higher than 100 °C. As a result, food is cooked in a much shorter time with less energy.



Inverter air conditioner

It regulates temperature by adjusting the speed of the compressor appropriately, instead of turning the compressor on and off periodically like a traditional air conditioner does. This saves a large amount of energy in starting/restarting the compressor.

OXFORD UNIVERSITY PRESS

Exercise 2.2

1 John replaces the lamp in his bedroom to make the room brighter. The new lamp must have a higher

A power rating.

B luminous flux.

C illuminance.

D efficacy.

(For Q2–3.) Four types of lamps are available in a shop (Table a).

	Power rating / W	Luminous flux / lm
P	30	1800
Q	40	2800
R	50	1600
S	60	3000

Table a

2 Which lamp uses the least amount of electricity after operating for 8 hours?

A *P* B *Q*

C *R* D *S*

3 Which lamp has the highest efficacy?

A *P* B *Q*

C *R* D *S*

(For Q4–5.) A torch emits a parallel light beam which falls on a plane surface as shown (Fig a).

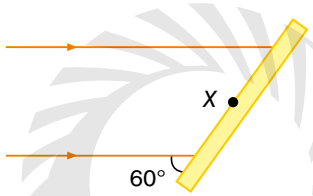


Fig a

4 The area of the illuminated region is 40 cm^2 and the illuminance at that region is 3000 lx. What is the luminous flux of the torch?

A 6 lm

B 12 lm

C 13.9 lm

D 75 lm

***5** Which of the following changes can increase the illuminance at *X*?

(1) Rotate the surface slightly clockwise about *X*.

(2) Rotate the surface slightly anticlockwise about *X*.

(3) Move the surface towards the left.

A (1) only

B (2) only

C (1) and (3) only

D (2) and (3) only

***6** Which of the following statements about the illuminance on a surface is/are correct?

(1) It is the total luminous flux received by the surface.

(2) It varies with the orientation of the surface.

(3) It is a value corrected for the dependence on wavelength of the sensitivity of the human eye.

A (1) only

B (2) only

C (1) and (3) only

D (2) and (3) only

7 Point *P* on a plane surface is directly below a point light source (Fig b). The illuminance at point *P* is *E*.

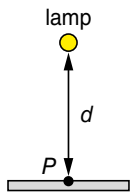
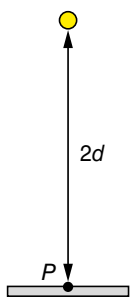


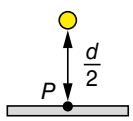
Fig b

In the following cases, express the illuminance at *P* in terms of *E*.

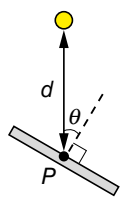
(a)



(b)



(c)



OXFORD UNIVERSITY PRESS

8 Figure c shows the information about a CFL printed on its package.

• Light Output	900 Lumens
• Colour Temperature	2700K WarmWhite
• Energy Used	15 Watts
• Lifetime	8000 Hours
• Base	E27 Screw Base
• Voltage	220-240V

Fig c

(a) What is the efficacy of the CFL?

(b) Electricity costs \$1.1 per kW h. The CFL is turned on for 4 hours each day. What is the cost of using the CFL for 365 days?

★**9** A lamp with a luminous flux of 6400 lm is placed at point X (Fig d). Find the illuminances at A and B. Assume the lamp is a point source.

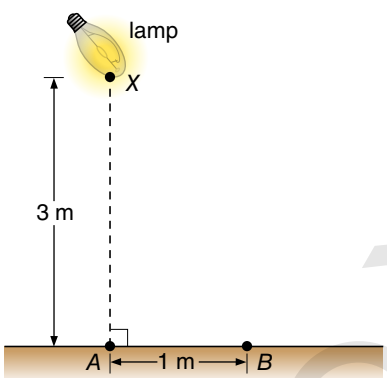


Fig d

★**10** Tommy is reading a newspaper which is slightly tilted towards him (Fig e). Light from a lamp hits the newspaper at an angle of 40° with the normal of the newspaper surface.

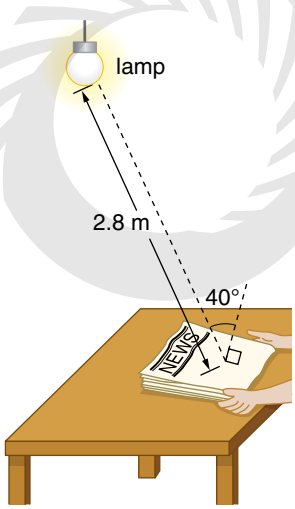


Fig e

(a) The illuminance on the newspaper due to the lamp is 11 lx. Estimate the luminous flux of the lamp. Assume the lamp is a point source.

(b) If Tommy lays the newspaper on the table horizontally, how will the illuminance on the newspaper change?

★**11** Figure f shows the energy of different wavelengths of light given out by lamps X and Y.

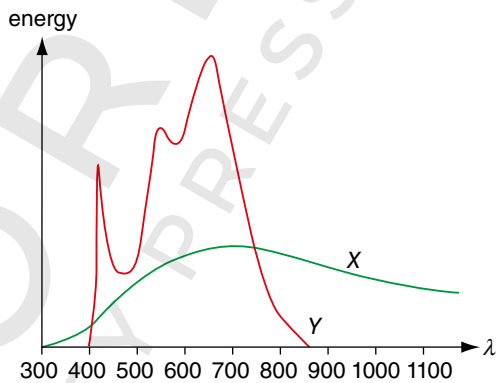


Fig f

If the output power of lamps X and Y are the same, which lamp will look brighter? Explain briefly.

★**12** The Earth is 1.5×10^{11} m away from the Sun. The sunlight reaching the earth appear parallel. At a certain instant, sunlight falls at 55° on city X (Fig g). The illuminance at X is 7.46×10^4 lx.

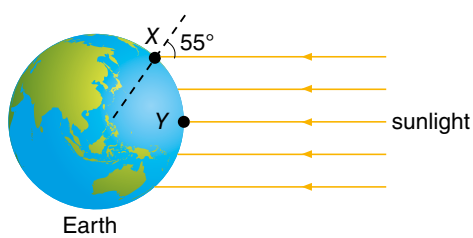


Fig g

(a) At the same instant, sunlight falls perpendicularly on city Y. Estimate the illuminance at Y.

(b) Estimate the luminous flux of the Sun.

OXFORD UNIVERSITY PRESS

Key concepts 2

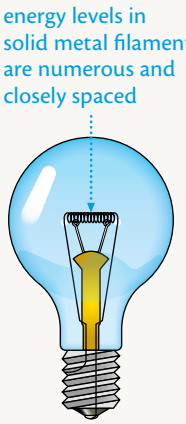
1 Electric light

Incandescent lamp

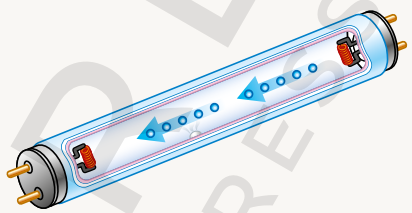
① Electric current heats the metal filament to around 2500 °C.

② Electrons in the filament are excited.

③ Excited electrons jump to different lower energy levels to release EM waves of a broad range of frequencies (only 5–10% is visible light).



Fluorescent tube lamp (FTL)



① Voltage is applied across the ends and current flows.

② The current heats up the tube and cause the mercury inside to vaporize.

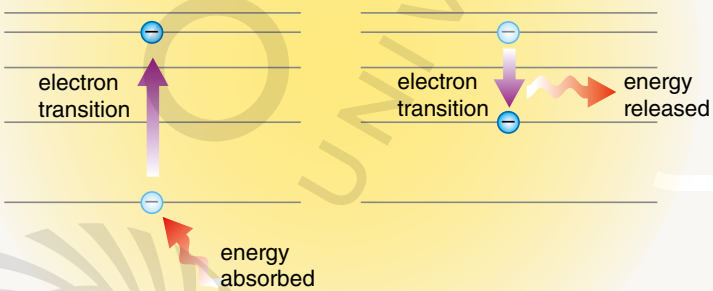
③ Electrons in mercury atoms are excited.

④ Excited electrons jump to lower energy levels to release UV.

⑤ Phosphor coating absorbs UV and emits visible light.

Electron transition

electron in atom jumps from one energy level to another by absorbing or releasing energy

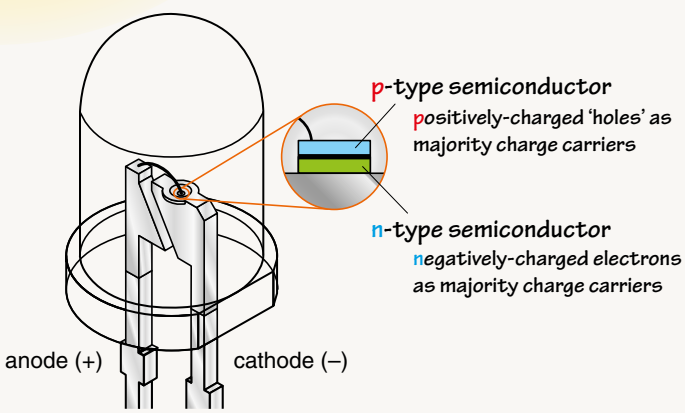


Light emitting diode (LED)

① A forward d.c. voltage is applied.

② Electrons and holes move towards the p-n junction and combine.

③ Electrons move to a lower energy level and emit EM wave (regarded as monochromatic).



2

How bright is a light source?

Luminous flux (Φ) of a light source

- power emitted by a light source, with correction for human vision
- unit: lumen (lm)



most sensitive to 555 nm (green-yellow)

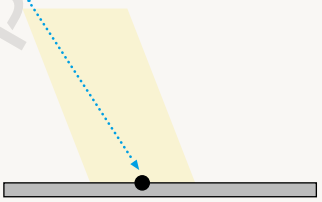
3

How much light is received by a surface?

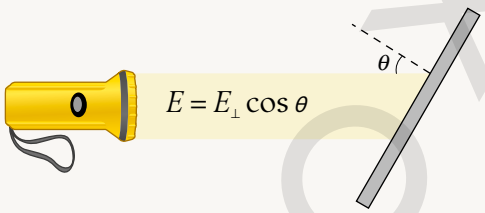
Illuminance (*E*) on a surface

- luminous flux falling on a surface per unit area
- unit: lux (lx)

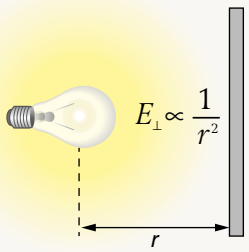
$$E = \frac{\Phi}{A}$$

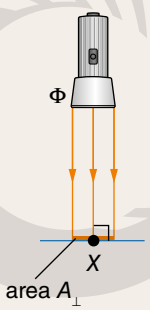
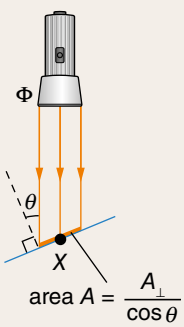
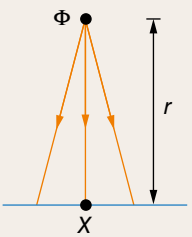
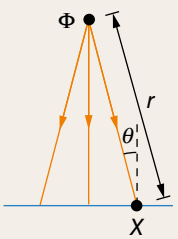


Lambert's cosine law



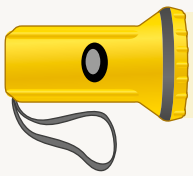
Inverse-square law



	Parallel source		Point source	
Cases				
E at X	$E_\perp = \frac{\Phi}{A_\perp}$	$E = E_\perp \cos\theta = \frac{\Phi \cos\theta}{A_\perp}$	$E_\perp = \frac{\Phi}{4\pi r^2}$	$E = E_\perp \cos\theta = \frac{\Phi \cos\theta}{4\pi r^2}$

OXFORD UNIVERSITY PRESS

4 How efficiently a light source uses energy for illumination



Efficacy (in lm W^{-1}) = $\frac{\text{luminous flux}}{\text{input power}}$

Key terms

1 compact fluorescent lamp (CFL) 緊湊型熒光燈 p.49
2 energy level 能級 p.46
3 fluorescent tube lamp (FTL) 熒光燈 p.49
4 gas discharge lamp 氣體放電燈 p.47
5 illuminance 照明度 / 光照度 p.59
6 incandescent lamp 白熾燈 p.47
7 inverse-square law 平方反比定律 p.63
8 Lambert's cosine law 朗伯餘弦定律 p.60
9 LED lamp 發光二極管燈 / LED 燈 p.47
10 light emitting diode (LED) 發光二極管 p.52
11 lumen (lm) 流明 p.57
12 luminous efficacy / efficacy 光視效能 p.69
13 luminous flux 光通量 p.57
14 lux (lx) 勒克司 p.59
15 semiconductor 半導體 p.52

Revision 2

Multiple-choice questions

1 2.2 Which of the following is the unit of illuminance?

A lm
B lm s^{-1}
C lm W^{-1}
D lm m^{-2}

***2** Misc Which of the following statements about an incandescent lamp is correct?

(1) Visible light is produced by the electron transitions occurring in its filament.
(2) Its luminous flux is generally lower than that of a compact fluorescent lamp.
(3) Its efficacy is generally higher than that of a compact fluorescent lamp.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

***3** 2.2

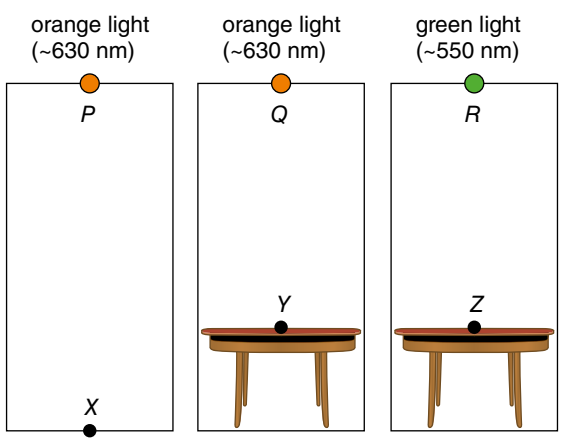


Fig a

Light bulbs *P*, *Q* and *R* emit monochromatic light of the same power (Fig a). Arrange the illuminance at *X*, *Y* and *Z* in ascending order.

A *X*, *Y*, *Z*
B *X*, *Z*, *Y*
C *Y*, *X*, *Z*
D *Z*, *Y*, *X*

OXFORD UNIVERSITY PRESS

★4 2.2 Lamp X has a higher efficacy than lamp Y. The two lamps have the same input power. Which of the following statements must be correct?

(1) Lamp X looks brighter than lamp Y.

(2) The cost of electricity in using lamp X is lower than lamp Y.

(3) The human eye is more sensitive to the light emitted by lamp X than that by lamp Y.

A (1) only B (2) only

C (1) and (2) only D (1), (2) and (3)

★5 2.2 Which of the following statements about the luminous flux of a point source of light is/are correct?

(1) It is inversely proportional to the square of the distance from the source.

(2) Its unit is lm.

(3) It is the total power of visible light emitted by the source.

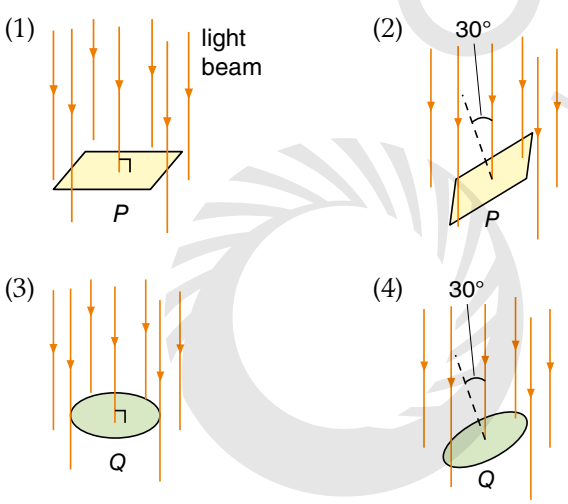
A (2) only

B (3) only

C (1) and (2) only

D (1) and (3) only

★6 2.2 Plane surfaces P and Q are put under a parallel light beam in different ways as shown. The area of P is larger than that of Q. In which case(s) is/are the illuminance on the surface the largest?



A (3) only

B (4) only

C (1) and (3) only

D (2) and (4) only

★7 2.1 Which of the following statements about an LED is/are correct?

(1) It emits EM waves of a broad range of frequencies.

(2) It can be operated by d.c. only.

(3) The frequencies of the EM waves emitted depend on the voltage across it.

A (1) only B (2) only

C (2) and (3) only D (1), (2) and (3)

★8 2.2 A photo on a wall faces a point source of light S as shown (Fig b). Incident light from S makes an angle of 60° with the normal at point P of the photo. The distance between P and S is r and the illuminance at P due to S is x.

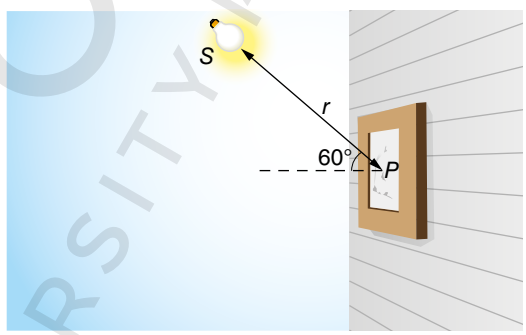


Fig b

If the distance between P and S is increased to $2r$ and light from S still hits P at 60°, what will the illuminance at P due to S become?

A $\frac{1}{8}x$ B $\frac{1}{4}x$

C $\frac{1}{2}x$ D x

★9 2.2 Lamps P and Q have the same luminous flux. P emits monochromatic red light while Q emits monochromatic green light. Which of the following statements must be correct?

(1) The efficacy of P is lower than that of Q.

(2) The input power of P is larger than that of Q.

(3) The useful output power of P is larger than that of Q.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

OXFORD UNIVERSITY PRESS

**10 2.2 A CFL, which can be considered as a point source, is located at a distance d from a plane surface (Fig c). The illuminance at point X on the surface is E. If the illuminance at Y is $2E$, what is θ?

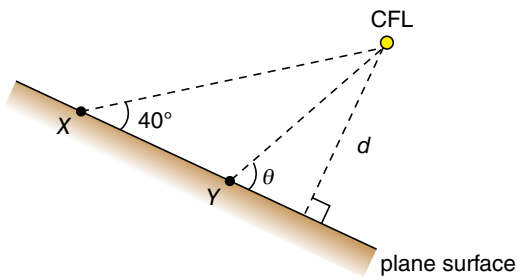


Fig c

A 15.2°

B 35.9°

C 54.1°

D 74.8°

Refer p.68

**11 2.2 A torch of luminous flux 15 lm emits a parallel light beam. A plane surface X of area 25 cm^2 blocks the light beam and a circular region of diameter 6 cm is illuminated (Fig d). If surface X is removed and another plane surface Y of area 2 cm^2 is placed to block part of the light beam (Fig e), what is the illuminance at point P on surface Y?

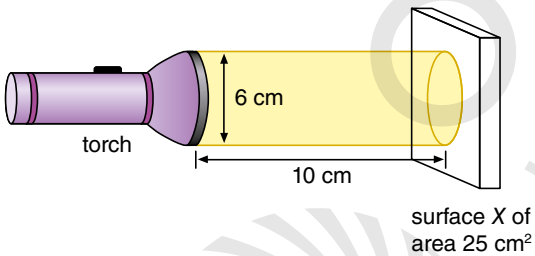


Fig d

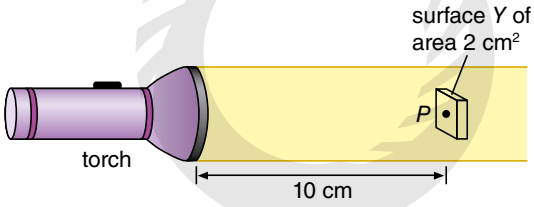


Fig e

A 119 lx

B 5310 lx

C 6000 lx

D 75 000 lx

Refer p.59

12 2.2 HKDSE 2015 Paper 2 Q3.1

A lamp is fixed on the ceiling of a room as the only light source (Fig f). The illuminance on the floor directly under the lamp is 600 lux. Assuming that the lamp emits light uniformly in all directions, what is the illuminance on the floor around point X as indicated in the diagram? Neglect any reflections of the walls and the ceiling.

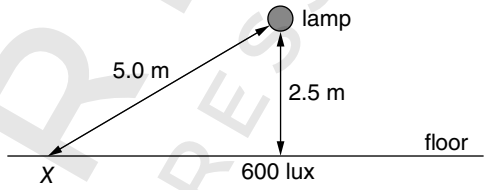


Fig f

A 75 lux (33%)

B 130 lux

C 150 lux

D 300 lux

13 2.2 HKDSE 2016 Paper 2 Q3.2

The signage below (Fig g) is to be illuminated by a point light source from the ground as shown (Fig h). What is the luminous flux of the source required for producing an illuminance of 200 lux at the centre of the signage? Assume that the source emits light uniformly in all directions and neglect any reflection from the ground.

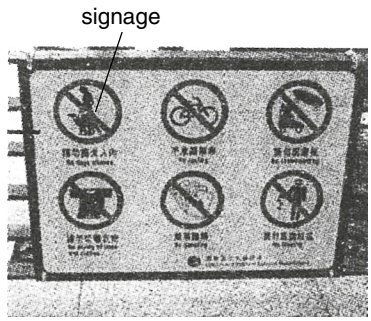


Fig g

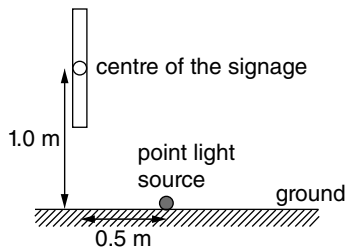


Fig h

A 7025 lm (26%)

B 3512 lm

C 3142 lm

D 560 lm

14 2.2 **HKDSE 2022 Paper 2 Q3.2**

Figure i shows a lamp of luminous flus 6000 lm placed 3 m above a horizontal mirror, and a book is placed horizontally at the same height as the lamp. The light from the lamp is reflected from the mirror to the book. If the mirror reflects 100% of the light energy, find the illuminance on the side of the book facing the mirror.

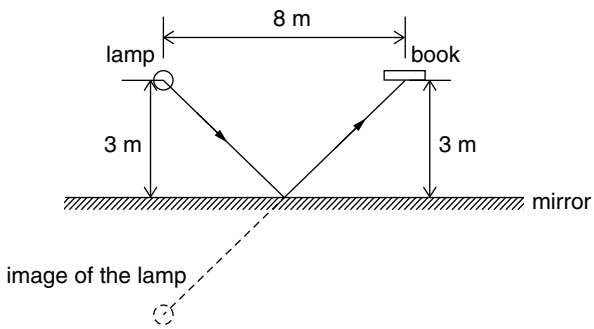


Fig i

A 2.9 lx (26%) B 3.8 lx

C 4.8 lx D 11.5 lx

Conventional questions

***15** Misc Tina wants to buy a lamp. The salesman tells her that an 8-W LED lamp appears as bright as a 40-W incandescent lamp (Fig j). The efficacy of the LED lamp is 60 lm W^{-1}.



Fig j

(a) What is the luminous flux of the LED lamp? (2 marks)

(b) What is the efficacy of the incandescent lamp? (1 mark)

(c) Tina uses the lamp for 5 hours a day on average. The cost of electricity is $1.1 per kW h. How much electricity cost did Tina saved per year if she uses the LED lamp? Take 1 year = 365 days. (2 marks)

(d) Besides energy efficiency, state another advantage of an LED lamp over an incandescent lamp. (1 mark)

***16** 2.2 Both the Earth and Mars are planets in the solar system. The distance of Mars from the Sun is about 1.5 times the distance of the Earth from the Sun (Fig k). At a certain time, sunlight falls perpendicularly on a city on the Earth, making the illuminance at the city to be 1.3×10^5 lx. Assume the sunlight is not blocked when travelling to the planets.

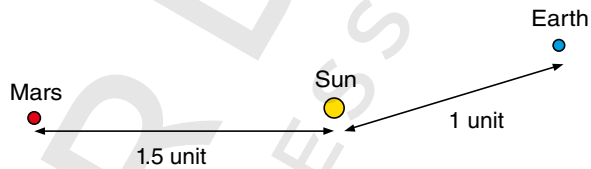


Fig k

(a) Estimate the illuminance at a point on Mars where sunlight falls perpendicularly on it. (2 marks)

(b) The Earth is 1.5×10^{11} m away from the Sun. Estimate the luminous flux of the Sun. (2 marks)

***17** Misc (a) Describe, in terms of electron transitions, the working principle of an incandescent lamp. (3 marks)

(b) Hence explain why the efficacy of the lamp is low. (1 mark)

(c) State an advantage and a disadvantage of an incandescent lamp when compared with a CFL. (2 marks)

***18** 2.2 Consider point X on a plane surface. A point light source has an input power of 60 W and is 2 m away from X as shown (Fig l). The illuminance at X is 50 lx.

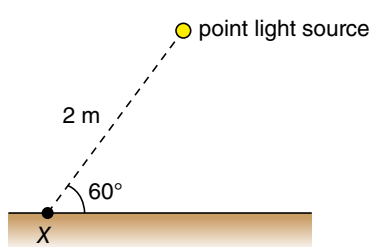


Fig l

(a) What is meant by illuminance? (1 mark)

(b) Find the luminous flux of the point light source. (2 marks)

(c) Find the efficacy of the point light source. (2 marks)

★19 Misc Figure m shows the energy label of a 20-W CFL.

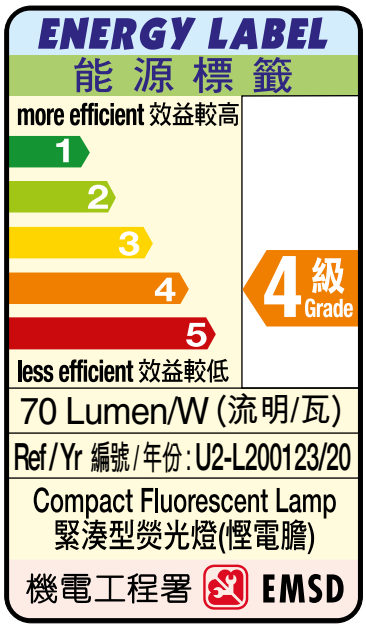


Fig m

(a) What is the luminous flux of the lamp? (2 marks)

(b) Bonnie installs the CFL on the ceiling of her room (Fig n). Assume the lamp is a point source. *P* is a point on the desk.

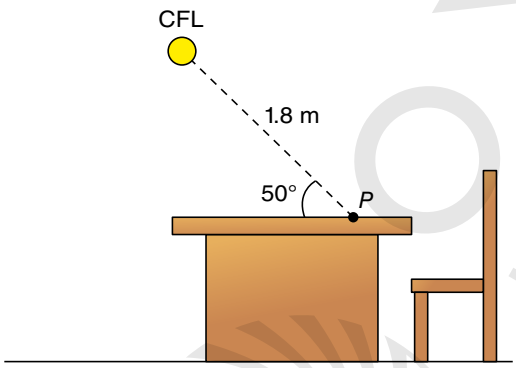


Fig n

(i) Estimate the illuminance at *P*. (2 marks)

(ii) Is the actual value of the illuminance at *P* larger than, equal to or smaller than, the value found in (i)? Explain briefly. (2 marks)

(c) The CFL has a special coating on its glass tube. Explain its function. (2 marks)

(d) An incandescent lamp of the same brightness consumes a much higher power. Explain why. (2 marks)

★20 Misc Mike is going to buy a lamp with a luminous flux greater than 1100 lm. He finds the following lamps in a shop (Table a).

Lamp	Power rating	Efficacy	Price
P	15 W	90 lm W^{-1}	$50
Q	60 W	17 lm W^{-1}	$30
R	100 W	12 lm W^{-1}	$10

Table a

(a) Show that lamp *Q* does not fulfil Mike's requirement. (1 mark)

(b) Consider lamps *P* and *R*. The cost of electricity is $1.1 per kW h. Find the number of days it takes for the difference in cost of electricity to compensate for the difference in price between the lamps if the lamps are used for 8 hours a day. (2 marks)

(c) Among lamps *P* and *R*, one is a compact fluorescent lamp (CFL) and the other is an incandescent lamp.

(i) Describe how visible light is produced by a CFL. (3 marks)

(ii) Explain which lamp is the CFL. (2 marks)

(d) Mike buys the CFL and installs it in his room (Fig o). Assume the lamp is a point source.

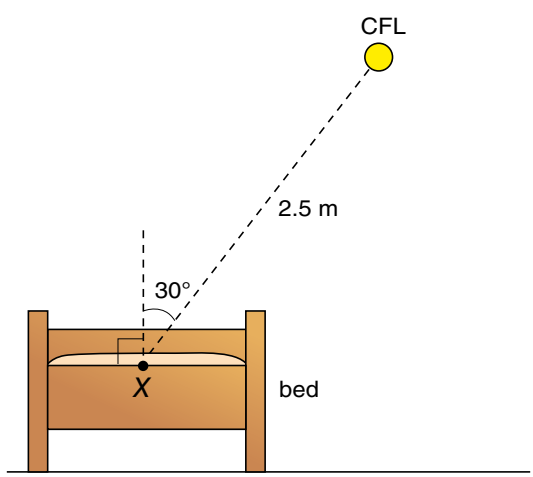


Fig o

X is a point on the surface of the mattress. The line connecting the lamp and *X* makes an angle of 30° with the normal at *X*. Find the illuminance at *X*. (2 marks)

**21 2.2 A student performs an experiment to estimate the luminous flux of a CFL (Fig p).

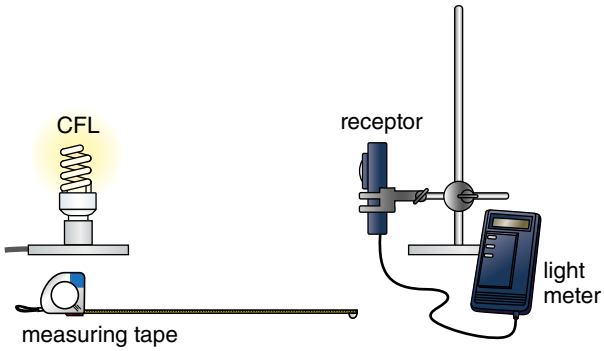


Fig p

The student uses a light meter to measure the illuminance E at different distances d from the CFL and the results are as follows.

d / m	1.0	1.1	1.2	1.3	1.4	1.5
E / lx	119	102	81	73	60	55

Table b

(a) Plot E against a suitable quantity to obtain a straight line graph which can demonstrate the inverse-square law. (4 marks)

(b) Estimate the luminous flux of the CFL from the graph in (a). (2 marks)

(c) Suggest two ways to improve the accuracy of this experiment. (2 marks)

Refer p.63–64

22 Misc HKDSE 2013 Paper 2 Q.3

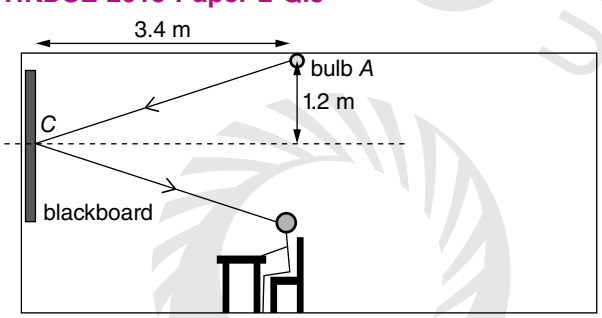


Fig q

The classroom shown in Figure q has an incandescent light bulb A of luminous flux 2000 lm (lumens). You may treat the light bulb as a point light source.

(a) Find the illuminance, in lm m^{-2}, by bulb A around the blackboard's centre C. Neglect any reflection of light. (2 marks)

(b) Bulb A is mainly for illuminating the student's desk, however, the light ray reflected back into student's eyes is undesirable (see the figure). Explain the type of surface that should be used for the blackboard so as to reduce such a problem. (2 marks)

(c) Figure r shows the appearance of the classroom. The average rate of heat gain of the classroom from outside is 14.5 kW.

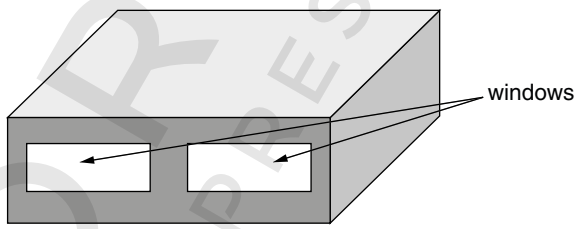


Fig r

The classroom is designed to accommodate a maximum of 15 persons at the same time and each person produces on average 100 J heat per second. There are altogether 6 identical incandescent light bulbs installed to illuminate the classroom and each bulb produces 80 J of heat per second.

(i) Estimate the cooling capacity, in kW, (due to heat produced inside the classroom and heat gain from outside) required for the classroom's air-conditioning system. Assume that there is no other equipment producing heat in the classroom. (2 marks)

(ii) The power rating of each light bulb is 100 W. the air-conditioning system consumes 0.5 J of electrical energy for removing 1 J of heat from the classroom. Estimate the total monthly cost of electricity for lighting and air-conditioning of the classroom operates 8 hours a day and 20 days a month. Given: cost of electricity = \$1.0 / kW h (3 marks)

(iii) Suggest one way of changing either the building structure or the electrical appliance so as to reduce the electricity bill through lower consumption of energy. (1 mark)

Q22 Exam Report: In (a), most candidates employed wrong formula/angle/distance or got sine and cosine mixed up. In (b), candidates were in general weak in their understanding of the concept of 'glare'. In (c)(i), many calculated the heat produced inside the room only. In (c)(ii), quite a number of the candidates wrongly stated 16.48 kW × 2 rather than 16.48 kW × 50%. Part (c)(iii) was well answered.

23 **HKDSE 2018 Paper 2 Q.3**

Misc

(a) (i) Incandescent lamps are far less energy efficient than other light sources like fluorescent lamps or light emitting diodes (LEDs). Explain why this is so in terms of how incandescent lamps produce light. (2 marks)

(ii) Two light sources of identical size and shape emit white light and green light respectively. If the light output power of them is the same, briefly explain which light source looks brighter. (2 marks)

(b) Figure s illustrates the simplified lighting arrangement within a tunnel in which two lamps each of luminous flux 10 000 lumens are installed on the ceiling.

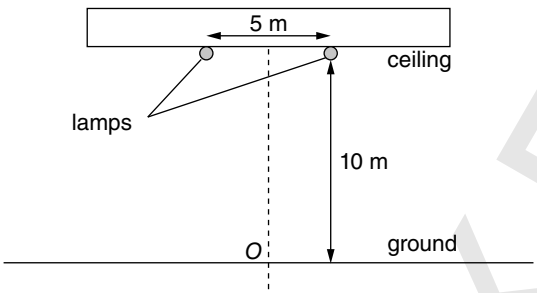


Fig s

(i) Calculate the illuminance around point O on the ground mid-way between the two lamps. Take the lamps as point light sources and reflection is assumed negligible. (3 marks)

Q23 Exam Report: In (a)(i), candidates in general knew that most of the energy became thermal energy in incandescent lamps, but many of them failed to state that these lamps produced light by joule heating of a tungsten filament at a high temperature. Candidates' performance in (a)(ii) was satisfactory though quite a number of them did not realise that luminous flux depends on the sensitivity of the human eye to light of different wavelengths. In (b)(i), most candidates were able to calculate the angle θ correctly, but some of them failed to apply $\frac{E \times \cos^3\theta \times 2}{4\pi d^2}$ or $\frac{E \times \cos\theta \times 2}{4\pi r^2}$ to find the illuminance. Part (b)(ii) was well answered. In (b)(iii), not many candidates were able to state explicitly the advantage and the disadvantage of the two arrangements of tunnel lights.

(ii) The specification of two kinds of lamps A and B are given below. In terms of efficacy recommend which one the tunnel company should choose. (1 mark)

lamp	rated power	luminous flux
A	150 W	11000 lumens
B	135 W	10000 lumens

Table c

(iii) Figure t and Figure u illustrate two arrangements of tunnel lights. In Figure t, more lamps each with relatively lower luminous flux are used. In Figure u, less lamps each with relatively higher luminous flux are used. The resultant average illuminance on the ground is the same in both cases.

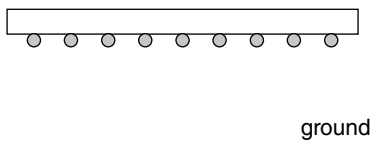


Fig t

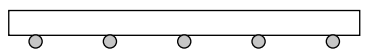


Fig u

State one advantage and one disadvantage of the arrangement in Figure t over that in Figure u. (2 marks)

OXFORD UNIVERSITY PRESS

Know more

Invention of the practical incandescent lamp (p.47)

The invention of the practical incandescent lamp is commonly credited to the American inventor Thomas Edison. In fact, the English physicist *Joseph Swan* invented the lamp (Fig a) and received a British patent in 1878, a year before Edison obtained his US patent for a similar lamp. Swan's house was the first place in the world to have light bulbs installed and operating. (Fig b)



Fig a *The first incandescent lamp invented by Swan.*

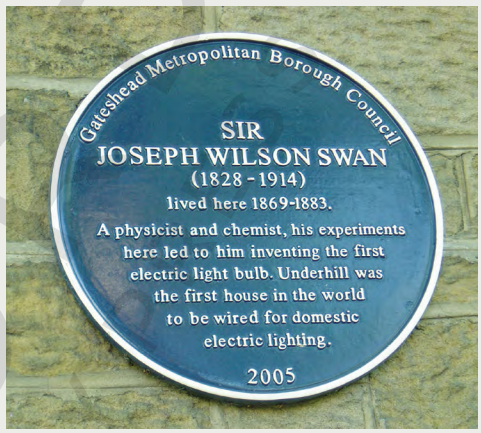


Fig b *Blue plaque at Swan's house.*

The world's longest-lasting incandescent lamp (p.48)

The Centennial Light (Fig c) in the USA is the world's longest-lasting incandescent lamp. It was first lit in 1901 and has operated for more than 1 000 000 hours, a thousand times longer than the average lifetime of a typical incandescent lamp nowadays. Its long life can be explained by its low operating power and nearly continuous operation.

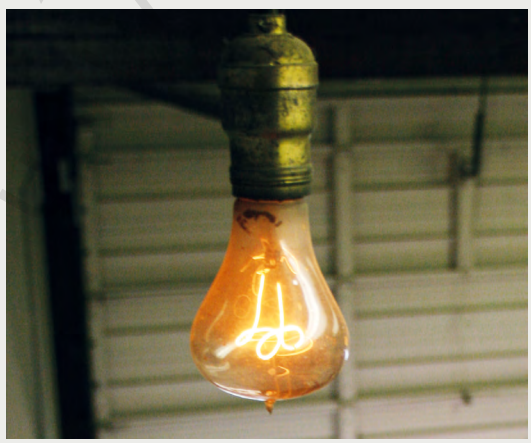


Fig c *The Centennial Light.*

Joseph Swan 斯旺